

MOHD TALIB

AI Engineering Lead · Full-Stack

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Engineering leader and de facto CTO with 5+ years building and leading cross-functional teams to ship production AI: multi-tenant agentic platforms, semantic search, and MLOps pipelines. As de facto CTO at Dumroo.ai, led a 20-engineer team to build Dumroo 2.0, now serving 6 K-12 districts (10+ schools), at 50% lower inference cost, 2x accuracy, and 10x faster onboarding. At TheAgentic AI, lead delivery of 10 enterprise platforms for 10 clients in 8 months and contribute to CortexON (455 GitHub stars). Led a 60+ engineer, 15-country team at Omdena. Author of the open-source agent framework OpenFishh.

EXPERIENCE

TheAgentic AI

Nov 2025 - Present

Tech Lead · Senior Full Stack AI Engineer

Applied AI Research, California · Remote

- *Lead engineer owning architecture and delivery of the company's enterprise AI platforms with a small forward-deployed team: client discovery, system design, and the path to production (2 to 4 weeks each).*
- **CortexON (open source).** Contributed core multi-agent task planning and execution APIs to TheAgentic's open-source generalized AI agent for everyday task automation. 455 GitHub stars.
- **Core platform engine.** Built a reusable, multi-tenant agentic orchestration engine (now the OpenFishh OSS framework) that shipped 10 production enterprise platforms for 10 clients in 8 months, with strict tenant isolation.
- **Enterprise peer-support platform.** Built a multi-tenant, privacy-first platform matching support seekers to mentors from voice stories: a safety-gated voice-intake pipeline (transcription, moderation and crisis classifier with human-in-the-loop, structured LLM extraction, multi-vector embeddings) feeding a learned-to-rank engine (pgvector ANN + LightGBM lambdarank that self-retrains from feedback at near-zero inference cost). FastAPI, Postgres/pgvector, async workers, Kubernetes. Enterprise and university pilots in 2026.
- **Reg-tech platform.** Replaced black-box RAG with deterministic SQL + LLM orchestration, making compliance workflows auditable and grounding every response in traceable CFR/FDA data.
- **Health-tech platform.** Built an async vision + LLM platform that generates metabolic insight reports from medical images on a scalable multi-tenant architecture.
- **Stakeholder coordination.** Ran weekly design reviews with enterprise client stakeholders, translating security and compliance constraints (FDA, CFR) into deterministic graph-based agent workflows.
- **Fundraising-ops platform.** Shipped new features and refactored an AI fundraising operating system used by founders across Google for Startups and NVIDIA Inception cohorts.

Dumroo.ai

Mar 2025 - Nov 2025

Tech Lead

AI EdTech, New Jersey · Remote

- *As de facto CTO reporting to the CEO, led a 20-engineer cross-functional team and owned hiring, developer training, sprint planning, and the architecture roadmap.*
- **Built Dumroo 2.0 from scratch in 4 weeks:** an AI learning and assessment platform now in production across US K-12 districts. Next.js, TypeScript, Supabase, Vercel, GitHub Actions.
- Built the core AI infrastructure: semantic vector caching and a dynamic LLM routing gateway that cut inference cost 50% while doubling accuracy.
- Partnered with school administrators and district heads to ship RBAC and automated onboarding pipelines, scaling adoption to 6 districts / 10+ schools in 6 months, with 10+ districts in the pipeline.
- Built internal developer-productivity and MLOps tooling (automated regression testing and evaluation pipelines) that cut setup and customer onboarding latency 10x.
- Integrated Stripe billing with role-based access; shipped CI/CD across Vercel and Netlify via GitHub Actions. Promoted to Senior in 6 months.

CodeSpaze

Apr 2024 - Feb 2025

Head of AI & Engineering (returned to lead)

Software, Web & AI Development · Remote

- *Returned to CodeSpaze to lead AI & Engineering after starting here as a Software Engineer in 2021.*
- Grew the AI Services division 3x in MRR by standardizing reusable ML templates and full-stack services.
- Led a 20-person cross-functional team (AI, backend, frontend, design, marketing) and set up CI/CD, automated testing, and developer onboarding to scale capacity for a pipeline of 10,000+ candidates.

- Built and deployed full-stack AI systems and internal automation tools, and ran technical training and mentoring for interns in Python, Data Science, and Web Development.

Techpile Technology

Jul 2023 - Apr 2024

Software Engineer & Data Analyst

Lucknow, India · On-site

- Built high-volume .NET Core APIs and tuned SQL schemas, cutting query latency 40% and improving dashboard load times. Earned Best Performer: Software Developer (2023).
- Led technical delivery of client web services and introduced reusable micro-frontends, winning project renewals and growing services revenue by delivering ahead of schedule.

Omdena

Mar 2023 - Jul 2023

Machine Learning Engineer to Data Engineering Lead (Contract)

Global AI for Social Good, New York · Remote

- Built and ran a semi-automated active-learning labeling pipeline using confidence-threshold pre-annotation, clearing a 100K+ healthcare dataset in 3 weeks and cutting the project timeline 30%.
- Led a 60+ ML engineer team across 15 countries to train, evaluate, and deploy YOLOv8 + PyTorch computer-vision pipelines, delivering the diagnostics system ahead of schedule.

CodeSpaze

Jul 2021 - Feb 2023

Software Engineer

Software, Web & AI Development · Remote

- Shipped 8+ client applications across web, AI, and automation, owning frontend and backend on 2 to 3 week cycles.
- Built reusable component libraries and internal tooling that cut project setup and delivery time by about 30%, and prototyped the early AI/ML work that seeded the company's AI service line.

SELECTED PROJECTS

OpenFishh | Python, FastAPI, PostgreSQL, pgvector, OpenAI, Anthropic, LangChain

 Live ·  Code

- Open-source agent infrastructure built around the three hardest problems in production agents: agent memory (persistent, provenance-tracked), agent reliability (verification, confidence scoring, safe fallbacks), and agent evaluation. The OSS counterpart to HumanizingLabs. Apache 2.0, 17 stars.

HumanizingLabs | Python, Multi-Agent AI, OpenAI, Anthropic, PostgreSQL, FastAPI, Redis, RAG

 Live

- A living society of 1,200 AI agents across 31 knowledge domains that reads the internet continuously, keeps persistent belief memory with full provenance, runs structured debates on contested claims, and produces auditable intelligence reports. Every belief is typed, sourced, and confidence-decomposed.

Wrively | Next.js, TypeScript, OpenAI, Supabase

 Live ·  Code

- AI writing platform that learns your voice and writes from it: topic to published post in under 3 minutes, with one-click multi-channel publish.

BlueDrum AI | React, FastAPI, AI/ML, PostgreSQL

 Live ·  Code

- Legal evidence platform that turns chats, documents, and financial records into structured case files lawyers can use, with AI classification, entity extraction, and timeline reconstruction.

TalPred AI | XGBoost, PostgreSQL, Redis, MLflow, React

 Code

- Production-grade equity alpha system with automated daily cross-sectional predictions, paper trading, monitoring, and calibrated backtesting with signal-decay analysis.

TECHNICAL SKILLS

AI Systems: Agent Orchestration, Agent Memory, Multi-Agent Systems, AI Reliability, Tool Calling, RAG, LLM Evaluation, Provenance & Observability, Fine-tuning, Prompt Engineering, LangChain, OpenAI / Anthropic / Gemini APIs, Vector DBs (pgvector), Vision-LLM

Full-Stack Engineering: TypeScript, Python, Next.js, React, FastAPI, Node.js, Express, Django, Flask, Tailwind CSS, React Native, .NET

Data & ML: PostgreSQL, pgvector, Supabase, Neon, Redis, MongoDB, MySQL, Firebase, SQLAlchemy, Alembic, MLflow, PyTorch, TensorFlow, scikit-learn, XGBoost, Computer Vision (YOLOv8, CNN)

Infrastructure & DevOps: Docker, AWS, Azure, Vercel, Netlify, Render, GitHub Actions, CI/CD, async architectures, REST API design, observability

Leadership & Process: Technical Leadership, Agile/Scrum, Sprint Planning, Hiring & Mentoring, Code Review, System Design, Stakeholder Management, 0-to-1 Delivery

EDUCATION

Sagar Institute of Technology & Management

B.Tech, Computer Science & Engineering

2019 - 2023

Barabanki, India

SELECTED COURSEWORK & CERTIFICATIONS

Deep Learning Specialization · DeepLearning.AI (Andrew Ng)

Machine Learning · Stanford University (Coursera)

Generative AI with Large Language Models · DeepLearning.AI and AWS

LangChain for LLM Application Development · DeepLearning.AI

Prompt Engineering for Developers · DeepLearning.AI

MLOps Specialization · DeepLearning.AI

RECOGNITION & AWARDS

- **Fastest promotion to Senior Engineer** · Dumroo.ai, 2025 (six months from Full Stack AI Engineer to Senior)
- **Apache 2.0 open-source author** · OpenFishh (17 stars, 3 forks)
- **Top 10, Data Science Hackathon** · Innomatics Research Labs, 2023
- **Best Performer, Software Developer** · Techpile Technology, 2023
- **Letters of Recommendation** · CodeClause and The Sparks Foundation